# DI WU (Assistant Professor, ECE, UCF)

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www.unarylab.com
https://scholar.google.com/citations?user=v6DNkTAAAAAJ

## APPOINTMENT

**Assistant Professor** | ***Electrical and Computer Engineering*** 08/2023 – Present
University of Central Florida
**Joint Assistant Professor** | ***Computer Science*** 08/2023 – Present
University of Central Florida

## RESEARCH INTEREST

- Computer architecture
  — performance, efficiency, etc.
- Emerging computing
  — unary, neuromorphic, quantum, etc.
- Domain specific acceleration
  — artificial intelligence, brain computer interface, etc.
- Heterogeneous system
  — Machine learning systems, multi-GPU systems, etc.

## EDUCATION

**Doctor of Philosophy** | ***Electrical and Computer Engineering*** 09/2017 – 07/2023
University of Wisconsin–Madison
- Advisor: Prof. Joshua San Miguel
- Thesis: Power-Efficient Computer Architecture via Unary and Approximate Computing
  Harold Peterson Outstanding Dissertation Award

**Master of Engineering** | ***Integrated Circuit Engineering*** 09/2012 – 01/2015
Fudan University
**Bachelor of Science** | ***Microelectronics*** 09/2007 – 07/2012
Fudan University

## EMPLOYMENT

**Research Assistant** 09/2017 – 07/2023
Department of Electrical and Computer Engineering, UW–Madison
**Research Intern** 05/2022 – 09/2022, 05/2020 – 09/2020
Cerebras Systems
**Research Intern** 05/2019 – 09/2019
Meta (Formerly Facebook)
**Digital Circuit Engineer** 03/2015 – 05/2017
HiSilicon
**Research Assistant** 09/2012 – 01/2015
State Key Laboratory of ASIC and System, Fudan University

## HONORS AND AWARDS

AMD Faculty Award 2026
**Amar Mukherjee Best Paper Award of ISVLSI** 2025
**IEEE Micro Top Pick Honorable Mention** 2025
AMD Faculty Award 2024
**Harold Peterson Outstanding Dissertation Award at UW–Madison** 2024
**Distinguished Artifact Evaluation Award of ASPLOS** 2024
MLCommons Machine Learning and Systems Rising Star 2023
Capstone PhD Teaching Award Nomination at UW–Madison 2022
Grainger Wisconsin Distinguished Graduate Fellowship at UW–Madison 2022
Ph.D. Forum of DAC 2021
**IEEE Micro Top Pick** 2021
Gerald Holdridge Outstanding Teaching Assistant Award at UW–Madison 2020
Chancellor's Opportunity Fellowship at UW–Madison 2019
Qualcomm Innovation Fellowship Finalist 2019
Foxconn SmartCity Competition Winner 2019
Hiran Mayukh Award at UW–Madison 2018
Rising Star Award at HiSilicon 2015
National Scholarship at Fudan University (1/67) 2015
Excellent Student Union Leader at Fudan University 2010
Third Prize Freshman Scholarship at Fudan University (3/45) 2007

## PUBLICATIONS

$^{+}$ – Student $^{*}$ – Collaborator $^{\times}$ – Advisor – Top tier – Award

### Conference

[1] Estimating The Energy Consumption of Quantum Computing from A Full System Aspect
Siyuan Niu, **Di Wu**, Ozgur Kilic, Kwangmin Yu
*International Green and Sustainable Computing Conference*, 2026

[2] Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs [acceptance rate: 19.1%]
Marco Kurzynski$^{+}$, Shaizeen Aga$^{*}$, **Di Wu**
*International Symposium on Computer Architecture*, 2026
Open-source software: Lit Silicon

[3] CryptOracle: A Modular Framework to Characterize FHE
Cory Brynds$^{+}$, Parker McLeod$^{+}$, Lauren Caccamise$^{+}$, Asmita Pal, Dewan Saiham, Sazadur Rahman$^{*}$, Joshua San Miguel$^{\times}$, **Di Wu**
*International Symposium on Performance Analysis of Systems and Software*, 2026, DOI: 10.1109/ISPASS69572.2026.00040
Open-source software: CryptOracle

[4] Mugi: Value Level Parallelism For Efficient LLMs [acceptance rate: 14.5%]
Daniel Price$^{+}$, Prabhu Vellaisamy, John Shen$^{*}$, **Di Wu**
*International Conference on Architectural Support for Programming Languages and Operating Systems*, 2026, DOI: 10.1145/3779212.3790189
Open-source software: artifact

[5] PIM-SUM: Fast and Reliable In-Memory Summation for Recommendation Systems
Fan Li, Ruizhi Zhu, Huize Li, **Di Wu**, Xin Xin$^{*}$
*International Conference on Computer Design*, 2025, DOI: 10.1109/ICCD65941.2025.00042

[6] Can Photonic Interconnects be used for High-Throughput Memory Access in FHE Accelerators?
Dewan Saiham, Mariam Rabadi, **Di Wu**, Sazadur Rahman$^{*}$
*International Symposium on Low Power Electronics and Design*, 2025, DOI: 10.1109/ISLPED65674.2025.11261747

[7] Catwalk: Unary Top-K for Efficient Ramp-No-Leak Neuron Design for Temporal Neural Networks
Devon Lister$^{+}$, Prabhu Vellaisamy, John Shen$^{*}$, **Di Wu**
*IEEE Computer Society Annual Symposium on VLSI*, 2025, DOI: 10.1109/ISVLSI65124.2025.11130314
Amar Mukherjee Best Paper Award

[8] Leveraging Photonic Interconnects for Scalable and Efficient Fully Homomorphic Encryption
Dewan Saiham, **Di Wu**, Sazadur Rahman$^{*}$
*Government Microcircuit Applications & Critical Technology Conference*, 2025

[9] LoAS: Fully Temporal-Parallel Dataflow for Dual-Sparse Spiking Neural Networks [acceptance rate: 22.7%]
Ruokai Yin, Youngeun Kim, **Di Wu**, Priyadarshini Panda$^{*}$
*International Symposium on Microarchitecture*, 2024, DOI: 10.1109/MICRO61859.2024.00084
Open-source software: artifact

[10] Exploration of Unary Arithmetic-Based Matrix Multiply Units for Low Precision DL Accelerators
Prabhu Vellaisamy, Harideep Nair, **Di Wu**, Shawn Blanton$^{*}$, John Paul Shen$^{*}$
*IEEE Computer Society Annual Symposium on VLSI*, 2024, DOI: 10.1109/ISVLSI61997.2024.00126

[11] ALISA: Accelerating Large Language Model Inference via Sparsity-Aware KV Caching [acceptance rate: 19.6%]
Youpeng Zhao, **Di Wu**, Jun Wang$^{*}$
*International Symposium on Computer Architecture*, 2024, DOI: 10.1109/ISCA59077.2024.00077

[12] Carat: Unlocking Value-Level Parallelism for Multiplier-Free GEMMs [acceptance rate: 20.9%]
Zhewen Pan, Joshua San Miguel$^{\times}$, **Di Wu**
*International Conference on Architectural Support for Programming Languages and Operating Systems*, 2024, DOI: 10.1145/3620665.3640364
IEEE Micro Top Pick Honorable Mention (top 24 from all computer architecture papers)
Distinguished Artifact Evaluation Award
Open-source software: artifact

[13] uBrain: A Unary Brain Computer Interface [acceptance rate: 16.6%]
**Di Wu**, Jingjie Li, Zhewen Pan, Younghyun Kim$^{*}$, Joshua San Miguel$^{\times}$
*International Symposium on Computer Architecture*, 2022, DOI: 10.1145/3470496.3527401

[14] uSystolic: Byte-Crawling Unary Systolic Array [acceptance rate: 30.5%]
**Di Wu**, Joshua San Miguel[×]
*International Symposium on High-Performance Computer Architecture*, 2022, DOI: 10.1109/HPCA53966.2022.00010
Open-source software: uSystolic-Sim

[15] When Dataflows Converge: Reconfigurable and Approximate Computing for Emerging Neural Networks
**Di Wu**, Joshua San Miguel[×]
*International Conference on Computer Design*, 2021, DOI: 10.1109/ICCD53106.2021.00014

[16] UNO: Virtualizing and Unifying Nonlinear Operations for Emerging Neural Networks
**Di Wu**, Jingjie Li, Setareh Behroozi, Younghyun Kim[*], Joshua San Miguel[×]
*International Symposium on Low Power Electronics and Design*, 2021, DOI: 10.1109/ISLPED52811.2021.9502473

[17] Normalized Stability: A Cross-Level Design Metric for Early Termination in Stochastic Computing
**Di Wu**, Ruokai Yin, Joshua San Miguel[×]
*Asia and South Pacific Design Automation Conference*, 2021, DOI: 10.1145/3394885.3431549

[18] uGEMM: Unary Computing Architecture for GEMM Applications [acceptance rate: 18.0%]
**Di Wu**, Jingjie Li, Ruokai Yin, Hsuan Hsiao, Younghyun Kim[*], Joshua San Miguel[×]
*International Symposium on Computer Architecture*, 2020, DOI: 10.1109/ISCA45697.2020.00040
IEEE Micro Top Pick (top 12 from all computer architecture papers)
Open-source software: UnarySim

[19] Approximate Hardware Techniques for Energy-Quality Scaling Across the System
Younghyun Kim[*], Joshua San Miguel[×], Setareh Behroozi, Tianen Chen, Kyuin Lee, Yongwoo Lee, Jingjie Li, **Di Wu**
*International Conference on Electronics, Information, and Communication*, 2020, DOI: 10.1109/ICEIC49074.2020.9051208

[20] SECO: A Scalable Accuracy Approximate Exponential Function Via Cross-Layer Optimization
**Di Wu**, Tianen Chen, Chienfu Chen, Oghenefego Ahia, Joshua San Miguel[×], Mikko Lipasti[*], Younghyun Kim[*]
*International Symposium on Low Power Electronics and Design*, 2019, DOI: 10.1109/ISLPED.2019.8824959

[21] In-Stream Stochastic Division and Square Root via Correlation [acceptance rate: 18.9%]
**Di Wu**, Joshua San Miguel[×]
*Design Automation Conference*, 2019, DOI: 10.1145/3316781.3317844

[22] Convergence-Optimized Variable Node Structure for Stochastic LDPC Decoder
Qichen Zhang, Yun Chen, **Di Wu**, Xiaoyang Zeng, Yeong-luh Ueng
*International Conference on Acoustics, Speech and Signal Processing*, 2016, DOI: 10.1109/ICASSP.2016.7472936

[23] An Area-Efficient Architecture for Stochastic LDPC Decoder
Qichen Zhang, Yun Chen, **Di Wu**, Xiaoyang Zeng, Yeong-luh Ueng
*International Conference on Digital Signal Processing*, 2015, DOI: 10.1109/ICDSP.2015.7251868

[24] Latency-Optimized Stochastic LDPC Decoder for High-Throughput Applications
**Di Wu**, Yun Chen, Qichen Zhang, Lirong Zheng, Xiaoyang Zeng, Yeong-luh Ueng
*International Symposium on Circuits and Systems*, 2015, DOI: 10.1109/ISCAS.2015.7169329

[25] A High-Throughput LDPC Decoder for Optical Communication
**Di Wu**, Yun Chen, Yuebin Huang, Yeongluh Ueng, Lirong Zheng, Xiaoyang Zeng
*International Conference on ASIC*, 2013, DOI: 10.1109/ASICON.2013.6811973

## Journal

[1] OptoLink: Breaking Memory Bandwidth Bottlenecks in FHE Accelerators with Photonic Interconnects
Dewan Saiham, **Di Wu**, Sazadur Rahman[*]
*ACM Transactions on Design Automation of Electronic Systems* (2026), DOI: https://doi.org/10.1145/3806395

[2] Toward Plug & Play Myoelectric Control via One-Shot Latent Representation Alignment [impact factor: 6.6]
Zubaidah Al-Mashhadani[+], Trevor Overton, **Di Wu**, Mohsen Rakhshan[*]
*IEEE Transactions on Neural Systems & Rehabilitation Engineering* 33 (2025), pp. 4409–4421, DOI: 10.1109/TNSRE.2025.3626255

[3] Synergizing Quantum Techniques with Machine Learning for Advancing Drug Discovery Challenge
Zhiding Liang, Zichang He, Yue Sun, Dylan Herman, Qingyue Jiao, Yanzhang Zhu[+], Weiwen Jiang[*], Xiaowei Xu[*], **Di Wu**, Marco Pistoia[*], Yiyu Shi[*]
*Scientific Reports* 14.1 (2024), p. 31216, DOI: 10.1038/s41598-024-82576-4

[4] uGEMM: Unary Computing for GEMM Applications
**Di Wu**, Jingjie Li, Ruokai Yin, Hsuan Hsiao, Younghyun Kim[*], Joshua San Miguel[×]
*IEEE Micro* 41.3 (2021), pp. 50–56, DOI: 10.1109/MM.2021.3065369
IEEE Micro Top Pick (top 12 from all computer architecture papers)

[5] In-Stream Correlation-Based Division and Bit-Inserting Square Root in Stochastic Computing
**Di Wu**, Ruokai Yin, Joshua San Miguel[×]
*IEEE Design & Test* 38.6 (2021), pp. 53–59, DOI: 10.1109/MDAT.2021.3050716

[6] Strategies for Reducing Decoding Cycles in Stochastic LDPC Decoders
**Di Wu**, Yun Chen, Qichen Zhang, Yeong-luh Ueng, Xiaoyang Zeng
*IEEE Transactions on Circuits and Systems II: Express Briefs* 63.9 (2016), pp. 873–877, DOI: 10.1109/TCSII.2016.2535038

[7] An Efficient Multirate LDPC-CC Decoder With a Layered Decoding Algorithm for the IEEE 1901 Standard
Yun Chen, Qichen Zhang, **Di Wu**, Changsheng Zhou, Xiaoyang Zeng
*IEEE Transactions on Circuits and Systems II: Express Briefs* 61.12 (2014), pp. 992–996, DOI: 10.1109/TCSII.2014.2362721

## Workshop and Poster

[1] PyTorch-Based Implementation of Relay-BP
Thomas Danielsen[+], Caden Brock[+], Yanzhang Zhu[+], **Di Wu**
*Florida Quantum Conference* (2026)

[2] Mugi: Value Level Parallelism For Nonlinear Operations in LLMs
Daniel Price[+], Prabhu Vellaisamy, John Shen[*], **Di Wu**
*Workshop on Unary Computing, collocated with ASPLOS* (2026)

[3] Agraph: A unified Graph Representation for At-Will Simulation of Emerging Stacks
Daniel Price[+], Prabhu Vellaisamy, Patricia Gonzalez[*], George Michelogiannakis[*], John Shen[*], **Di Wu**
*Workshop on Unary Computing, collocated with ASPLOS* (2026)

[4] Sense-as-You-Go: A Neuromorphic Framework for Efficient Edge Sensing and Processing
**Di Wu**, Mohsen Rakhshan[*]
*Energy Consequences of Information Workshop* (2026)

[5] A-Graph: A Unified System Representation for Automated Cross-Stack Optimization
**Di Wu**
*Energy Consequences of Information Workshop* (2026)

[6] Catwalk Neuron for Efficient Temporal Neural Networks
**Di Wu**
*Energy Consequences of Information Workshop* (2026)

[7] EnerQy: Energy Estimation for Quantum Computing
**Di Wu**, Siyuan Niu[*], Zhipeng Deng[*]
*Energy Consequences of Information Workshop* (2026)

[8] Syndrilla: Simulating Decoders for Quantum Error Correction using PyTorch
Yanzhang Zhu[+], Chen-Yu Peng, Yun Hao Chen, Siyuan Niu[*], Yeong-Luh Ueng[*], **Di Wu**
*IEEE International Conference on Quantum Computing and Engineering*, 2025, DOI: 10.1109/QCE65121.2025.10470
Open-source software: Syndrilla

[9] Are We Scaling the Right Thing? A System Perspective on Test-Time Scaling
Youpeng Zhao, Jinpeng LV[*], **Di Wu**, Jun Wang[*], Christopher Gooley[*]
*NeurIPS Workshop on Efficient Reasoning* (2025)

[10] Synergizing Error Suppression, Mitigation and Correction for Fault-Tolerant Quantum Computing
Yanzhang Zhu[+], Siyuan Niu[*], **Di Wu**
*Workshop on Quantum Intelligence, Learning & Security, collocated with IEEE TPS* (2024), DOI: 10.1109/TPS-ISA62245.2024.00065

[11] Exploration of Unary Arithmetic-Based Matrix Multiply Units for Low Precision DL Accelerators
Prabhu Vellaisamy, Harideep Nair, **Di Wu**, Shawn Blanton[*], John Paul Shen[*]
*Workshop on Unary Computing, collocated with ASPLOS* (2024)

[12] xBrain: Brain-Like Computing for Explainable Brain-Computer Interfaces
Queenly Xie[+], Prabhu Vellaisamy, **Di Wu**
*Young Architect Workshop, collocated with ASPLOS* (2024)

[13] T-MAC: Temporal Multiplication with Accumulation
Zhewen Pan, **Di Wu**, Joshua San Miguel[×]
*Young Architect Workshop, collocated with ASPLOS* (2022)

## Pre-Print and Under-Submission

[1] Speculative Matching for Real-Time Surface Code Decoding
Sumeet Shirgure, Yanzhang Zhu[+], **Di Wu**, Siyuan Niu[*]
*arXiv Pre-print* (2026)

[2] Can AI Review Improve Paper Drafting? An Empirical Study on 20 Computer Architecture Submissions
**Di Wu**
*arXiv Pre-print* (2026)
Open-source software: AI-Paper-Review

[3] Lottery BP: Unlocking Quantum Error Decoding at Scale
Yanzhang Zhu[+], Chen-Yu Peng, Yun Hao Chen, Yeong-Luh Ueng[*], **Di Wu**
*arXiv Pre-print (under review)* (2026)
Open-source software: Syndrilla

[4] uSense: Unary-Computing-based Stochastic Edge Neuromorphic Sensing
Zubaidah Al-Mashhadani[+], Mohsen Rakhshan[*], **Di Wu**
*Research Square Pre-print (under review)* (2026)

[5] A-Graph: A Unified Graph Representation for Cross-Stack Cost Modeling
Daniel Price[+], Prabhu Vellaisamy, Patricia Gonzalez[*], George Michelogiannakis[*], John Shen[*], **Di Wu**
*arXiv Pre-print (under review)* (2026)
Open-source software: Archx

[6] Chopper: A Multi-Level GPU Characterization Tool & Derived Insights Into LLM Training Inefficiency
Marco Kurzynski[+], Shaizeen Aga[*], **Di Wu**
*arXiv Pre-print* (2025)
Open-source software: Chopper

[7] Unleashing The Potential of LLMs for Quantum Computing: A Study in Quantum Architecture Design
Zhiding Liang, Jinglei Cheng, Rui Yang, Hang Ren, Zhixin Song, **Di Wu**, Tongyang Li[*], Yiyu Shi[*]
*arXiv Pre-print* (2023)

[8] Representation Range Needs for 16-Bit Neural Network Training
Valentina Popescu[*], Abhinav Venigalla[*], **Di Wu**, Robert Schreiber[*]
*arXiv Pre-print* (2021)
Open-source software: production-level cbfloat16

## Patent

[1] Computer Architecture with Value-Level Parallelism
Joshua San Miguel, Di Wu, Zhewen Pan
US Patent Application 18/221,127, 2025

## Media Coverage

**From Control to Data to Value: A Third Axis of Parallelism** 05/2026
ACM SIGARCH Computer Architecture Today

## Invited Talks

**Quantum Error Decoding At Scale with Lottery Belief Propagation**
Quantum Computing System Lecture Series 10/2026
**Lit Silicon: How Thermal Variation Impairs AI Performance via Concurrent Execution in GPU Nodes and Its Simple Repairs**
AMD Research and Advanced Development 05/2026
**Value-Level Parallelism: New Opportunities for Parallel Computing from Fewer Bits in the AI Era**
University of Pittsburgh 03/2026
University of Wisconsin–Madison 12/2025
Lawrence Berkeley National Laboratory 10/2025
NVIDIA Research 10/2025
**The Phantom of the Datacenter: Unmasking the Culprit behind Performance Variation in GenAI Training**
AMD Research and Advanced Development 10/2025
**Salvage Deep Learning Efficiency: A Unary Computing Approach**
University of California, Santa Cruz 02/2025
ShanghaiTech University 12/2024
Fudan University 12/2024
Case Western Reserve University 10/2024
Peking University 08/2024
University of Minnesota Twin Cities 03/2024
University of Louisiana at Lafayette 11/2023
**Unary Computing for Power-Efficient Computer Architecture**
AMD Research and Advanced Development 07/2023
University of Central Florida 02/2023
University of California, Los Angeles 11/2022
University of California, Santa Barbara 10/2022

## Teaching and Mentoring

**Instructor**

| | |
|---|---|
| EEL4768 (Computer Architecture), UCF | FA 2026 |
| EEL5796 (Big Data Computer Architecture and Systems), UCF | SP 2026 |
| EEL4768 (Computer Architecture), UCF | SP 2026 |
| EEE3342C (Digital Systems), UCF | FA 2025 |
| EEL4742C (Embedded Systems), UCF | SP 2025 |
| EEE3342C (Digital Systems), UCF | FA 2024 |
| EEL5796 (Big Data Computer Architecture and Systems), UCF | SP 2024 |
| ECE697 (Capstone Project in Machine Learning and Signal Processing), UW–Madison | SU 2023 |

**Teaching Assistant**

| | |
|---|---|
| ECE554 (Digital Engineering Lab), UW–Madison | SP 2022 |
| ECE454 (Mobile Computing Lab), UW–Madison | FA 2021 |
| ECE454 (Mobile Computing Lab), UW–Madison | FA 2020 |
| ECE554 (Digital Engineering Lab), UW–Madison | SP 2020 |
| ECE554 (Digital Engineering Lab), UW–Madison | FA 2019 |
| ECE554 (Digital Engineering Lab), UW–Madison | SP 2019 |
| ECE554 (Digital Engineering Lab), UW–Madison | FA 2018 |
| ECE552 (Introduction to Computer Architecture), UW–Madison | FA 2018 |

**Guest Lecturer**

| | |
|---|---|
| ECE18743 (Neuromorphic Computer Architecture & Processor Design), CMU | SP 2025 |
| ECE757 (Advanced Computer Architecture II), UW–Madison | SP 2021 |
| ECE757 (Advanced Computer Architecture II), UW–Madison | SP 2020 |
| ECE752 (Advanced Computer Architecture I), UW–Madison | SP 2019 |

## Professional Service

**Panelist**

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|---|---|
| NSF Medium Panel in Division of Computer and Network Systems (CNS) | 2024 |
| IEEE Workshop on Quantum Intelligence, Learning & Security (QUILLS) | 2024 |

**Committee**

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|---|---|
| Organizing Chair of Artifact Evaluation at ISPASS | 2026 |
| Organizing Chair of Workshop on Architecting Error Corrected Quantum Computers (ARQTEC) at HPCA | 2025 |
| Organizing Chair of Undergrad Panel on "Charging STEM Career" at UCF | 2024 |
| Organizing Chair of Workshop on Unary Computing (WUC) at ASPLOS | 2024, 2026 |
| Organizing Committee of Quantum Computing for Drug Discovery Challenge at ICCAD | 2023 |
| Program Committee of ASPLOS | 2026 |
| Program Committee of HPCA | 2024, 2025, 2026, 2027 |
| Program Committee of ISCA | 2024, 2025, 2026 |
| Program Committee of MICRO | 2025, 2026 |
| Program Committee of ICS | 2026 |
| Program Committee of IISWC | 2023, 2025 |
| Program Committee of ISPASS | 2024, 2025, 2026 |
| Program Committee of IPDPS | 2025 |
| Program Committee of DAC | 2025 |
| Program Committee of ICCAD | 2024 |
| Program Committee of DAC PhD Forum | 2024 |
| Program Committee of ICCD | 2023, 2024 |
| Program Committee of GLSVLSI | 2026 |
| Program Committee of ICRC | 2024, 2026 |
| Program Committee of ICA3PP | 2023 |
| Program Committee of IEEE Workshop on Quantum Intelligence, Learning & Security (QUILLS) | 2024 |
| Program Committee of Young Architect Workshop (YArch) | 2023, 2024, 2025, 2026 |
| Program Committee of ASPLOS Artifact Evaluation | 2020, 2021 |
| Program Committee of MICRO Artifact Evaluation | 2021 |

**Mentor**

| | |
|---|---|
| UCF IEEE Engineering in Medicine and Biology Society (EMBS) Student Branch Chapter | 2024 |
| Computer Architecture Long-term Mentoring (CALM) | 2024 |
| Young Architect Workshop (YArch) | 2023 |
| Undergrad Architecture Mentoring Workshop (uArch) | 2023, 2024 |
| MICRO – "Meet a Senior PhD Student" | 2020 |

**Journal Reviewer**

ACM Transactions on Architecture and Code Optimization (TACO)
ACM Transactions on Embedded Computing Systems (TECS)
ACM Transactions on Reconfigurable Technology and Systems (TRETS)
IEEE Computer Architecture Letter (CAL)
IEEE Transactions on Circuits and Systems I (TCAS-I)
IEEE Transactions on Computers (TC)
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
IEEE Transactions on Emerging Topics in Computing (TETC)
IEEE Transactions on Very Large Scale Integration (VLSI) Systems (TVLSI)
IEEE Signal Processing Letters (SPL)
Journal of Network and Computer Applications (JNCA)
npj Quantum Information

npj Unconventional Computing
**UCF Computer Architecture Seminar Series (publicly accessible)** 2023 – Present

## FUNDING

**AMD Faculty Award: Understanding and Optimizing Agentic AI Systems on AMD Platforms** 2026
PI: $100k (unrestricted gift)
**DOE: APT-Q: Algorithm-Architecture Co-design in Partial Fault-Tolerance for Quantum Utility** 2026
Co-PI, $180k/$400k
**NSF EAGER: SENSE: National Security Evaluation of Neurotechnology Systems and Emerging Tools** 2025
Co-PI: $100k/$300k
**AMD Faculty Award: Understanding Cutting-Edge Techniques for Large Language Model Serving on AMD Hardware Platforms** 2024
PI: $100k (unrestricted gift)
**AMD AI & HPC Cluster Award** 2024, 2025, 2026
**AMD Developer Cloud Credit: $2k** 2025
**NSF ACCESS Program** 2024

## STUDENTS

**Current PhD Students**
Yang Yu 2026 – Present
Chetan Choudhary 2025 – Present
Daniel Price 2024 – Present
Marco Kurzynski 2024 – Present
Yanzhang Zhu 2024 – Present
Zubaidah Al-Mashhadani 2024 – Present
**Former Students**
Devon Lister (BS@UCF) 2023 – 2026
Cory Brynds (BS/MS@UCF → full-time@AMD) 2023 – 2026
Lauren Caccamise (BS@UCF → PhD@Purdue) 2024 – 2025
Parker McLeod (BS@UCF → full-time@AMD) 2023 – 2024
Tyler Goldsmith (BS@UCF → full-time@AMD) 2023 – 2024
Mustafa Nisar (BS@UoT → co-op@AMD) 2023 – 2024
Zhewen Pan (MS@UW–Madison → PhD@UW–Madison) 2020 – 2022
Ruokai Yin (BS@UW–Madison → PhD@Yale) 2019 – 2021